

```

    "changed": false,
    "ping": "pong"
  }
$ ansible -i inventory/kvm/inventory nginx2 -m ping
nginx2 | SUCCESS => {
    "changed": false,
    "ping": "pong"
  }

$ ansible -i inventory/kvm/inventory web -m ping
nginx1 | SUCCESS => {
    "changed": false,
    "ping": "pong"
  }
nginx2 | SUCCESS => {
    "changed": false,
    "ping": "pong"
  }
}

```

Nginx

We will first set up the Nginx Web servers. The EPEL release RPM is installed and the HAProxy server IP address is added to `/etc/hosts` file on the instances. The YUM package manager is used to install Nginx, and then Port 80 is allowed through the firewall. We then start the Nginx Web server and wait for the server to listen on port 80. The Ansible playbook to install and set up Nginx is as follows:

```

---
- name: Setup Nginx web server
  hosts: web
  become: yes
  become_method: sudo
  gather_facts: yes
  tags: [web]

  tasks:
    - name: Install EPEL Release
      package:
        name: epel-release
        state: present

    - name: Add HAProxy server to /etc/hosts
      lineinfile:
        path: /etc/hosts
        line: "{{ hostvars['haproxy'].ansible_host }} haproxy"
        state: present

    - name: Install Nginx
      package:
        name: nginx
        state: present

    - name: Allow port 80

```

```

    shell: iptables -I INPUT -p tcp --dport 80 -m state
    --state NEW,ESTABLISHED -j ACCEPT

```

```

- name: Start Nginx server
  systemd:
    name: nginx
    enabled: yes
    state: started

- name: Wait for server to start
  wait_for:
    port: 80

```

The above playbook can be invoked using the following command:

```

$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/frontend.yml --tags web -vv -K

```

The `-vv` represents the verbosity in the Ansible output. You can use up to four 'v' s for a more detailed output. The `-K` option prompts for the sudo password for the guest CentOS user account.

You can now open the Nginx Web server URLs (<http://192.168.122.248> and <http://192.168.122.147>) in a browser to see the default Nginx home page as shown in Figure 1.

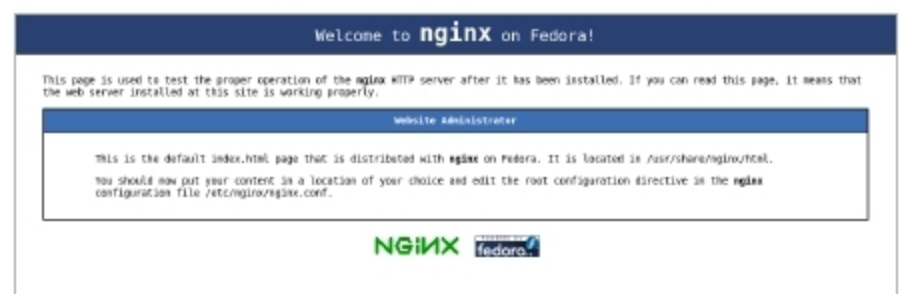


Figure 1: Default Nginx home page

HAProxy

The YUM package repository needs to be updated before proceeding to install HAProxy. The Nginx server IP addresses and hostnames are added to the `/etc/hosts` file on the instances. The default `/etc/haproxy/haproxy.cfg` directory is backed up and a new `haproxy.cfg` file is created, whose file contents are shown below:

```

global
    log                  127.0.0.1 local2

    chroot               /var/lib/haproxy
    pidfile               /var/run/haproxy.pid
    maxconn               4000
    user                  haproxy
    group                 haproxy
    daemon

    stats socket /var/lib/haproxy/stats

```